

Augustinos D. Saravanos

Postdoctoral Associate

Laboratory for Information & Decision Systems (LIDS)

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Citizenship: US, Greek

RESEARCH INTERESTS

The Science of Scaling Decisions: At the Intersection of Optimization and AI

Theory: large-scale optimization, optimal control, ML for optimization, generative modeling

Applications: multi-agent autonomy, robotics, transportation, logistics

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Ph.D. in Machine Learning

2025

Thesis: Distributed Optimization Architectures for Large-Scale Decision Making

Advisor: Evangelos A. Theodorou

Committee: Arkadi S. Nemirovski, Yao Xie, Justin Romberg, Efstathios Bakolas

Onassis Foundation Doctoral Fellow, 2021–2025

Georgia Institute of Technology

Atlanta, GA

M.Sc. in Aerospace Engineering

2024

University of Patras

Patras, Greece

Diploma in Electrical and Computer Engineering

2019

Thesis: Nonlinear Model Predictive Control for Space Robotic Systems

Co-advisors: Evangelos Papadopoulos, Nick Koussoulas

Class Rank: 2/211 (Top 1%)

EXPERIENCE

Massachusetts Institute of Technology (MIT)

Cambridge, MA

Postdoctoral Associate

Aug 2025-present

Supervisor: Chuchu Fan

Focus: Generative AI for optimization, decision-making and autonomy

Selected outcomes: ICML '26 Spotlight (top 2%), CoRL '26 Spotlight (top 3%); lead proposal author on two awarded industry-sponsored projects totaling \$370K.

BOSCH Center for Artificial Intelligence

Pittsburgh, PA

Machine Learning Research Intern

May-Aug 2023

Supervisor: Wan-Yi Lin

Focus: Collaborative learning via model alignment

Outcome: Three U.S. patent applications

Journal Publications

- [J4] [Second-Order Constrained Dynamic Optimization](#),
Y. Aoyama, O. So, **A.D. Saravanos** and E.A. Theodorou,
International Journal of Robotics Research (IJRR), 2026.
- [J3] [Distributed Covariance Steering via Non-Convex ADMM for Large-Scale Multi-Agent Systems](#),
A.D. Saravanos, I.M. Balci*, A.T. Abdul*, E. Bakolas and E.A. Theodorou,
IEEE Transactions on Automatic Control (TAC), 2026. Submitted.
- [J2] [Scaling Robust Optimization for Swarms: A Distributed Perspective](#),
A.T. Abdul*, **A.D. Saravanos*** and E.A. Theodorou,
IEEE Transactions on Automatic Control (TAC), 2026. Submitted.
- [J1] [Distributed Differential Dynamic Programming Architectures for Large-Scale Multi-Agent Control](#),
A.D. Saravanos, Y. Aoyama, H. Zhu and E.A. Theodorou,
IEEE Transactions on Robotics (T-RO), 2023.

Conference Publications

- [C18] [Learning to Optimize Actions with Flows](#),
A.D. Saravanos, O. So and C. Fan,
International Conference on Learning Representations (ICLR), 2027. Submitted.
- [C17] [Pareto-Optimal Entropy-Regularized Trajectory Optimization](#),
D.S. Georgiou, **A.D. Saravanos** and E.A. Theodorou
IEEE International Conference on Robotics and Automation (ICRA), 2027. Submitted.
- [C16] [Asynchronous Distributed Multi-Robot Motion Planning Under Imperfect Communication](#),
A. Tajbakhsh, **A.D. Saravanos**, J. Zhu, E.A. Theodorou, L.T. Biegler and A.M. Johnson,
IEEE International Conference on Robotics and Automation (ICRA), 2027. Submitted.
- [C15] [Language Grounded Bayesian Active Learning in Interactive Robot Planning](#),
H. Li, C. Sobolewski, **A.D. Saravanos**, W. Tan, J. Karigiannis and C. Fan,
Conference on Robot Learning (CoRL), 2026. **Spotlight (top 3%)**.
- [C14] [Learning-to-Optimize via Deep Unfolded Flows](#),
A.D. Saravanos, O. So, H.M. Sabbir Ahmad and C. Fan,
International Conference on Machine Learning (ICML), 2026. **Spotlight (top 2%)**.
- [C13] [Deep FlexQP: Accelerated Nonlinear Programming via Deep Unfolding](#),
A. Oshin, R. Ghosh, **A.D. Saravanos** and E.A. Theodorou,
International Conference on Learning Representations (ICLR), 2026.
- [C12] [Momentum Multi-Marginal Schrödinger Bridge Matching](#),
P. Theodoropoulos, **A.D. Saravanos**, E.A. Theodorou† and G.H. Liu†,
Neural Information Processing Systems (NeurIPS) 2025.
- [C11] [Nonlinear Robust Optimization for Planning and Control](#),
A.T. Abdul, **A.D. Saravanos** and E.A. Theodorou,
IEEE Conference on Decision and Control (CDC), 2025.
- [C10] [Operator Splitting Covariance Steering for Safe Stochastic Nonlinear Control](#),
A. Ratheesh, V. Pacelli, **A.D. Saravanos** and E.A. Theodorou,
IEEE Conference on Decision and Control (CDC), 2025.

- [C9] [Deep Distributed Optimization for Large-Scale Quadratic Programming](#),
A.D. Saravanos, H. Kuperman, A. Oshin, A.T. Abdul, V. Pacelli and E.A. Theodorou,
International Conference on Learning Representations (ICLR), 2025.
- [C8] [Scalable Robust Optimization for Safe Multi-Agent Control Under Deterministic Uncertainty](#),
A.T. Abdul*, **A.D. Saravanos*** and E.A. Theodorou,
American Control Conference (ACC), 2025.
- [C7] [Distributed Model Predictive Covariance Steering](#),
A.D. Saravanos, I.M. Balci, E. Bakolas and E.A. Theodorou,
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.
- [C6] [A Robust Differential Neural ODE Optimizer](#),
P. Theodoropoulos, G.H. Liu, T. Chen, **A.D. Saravanos** and E.A. Theodorou,
International Conference on Learning Representations (ICLR), 2024.
- [C5] [Distributed Hierarchical Distribution Control for Very-Large-Scale Clustered Multi-Agent Systems](#),
A.D. Saravanos, Y. Li and E.A. Theodorou,
Robotics: Science and Systems (RSS), 2023.
- [C4] [Improved Exploration for Safety-Embedded DDP Using Tolerant Barrier States](#),
J.E. Kuperman, H. Almubarak, **A.D. Saravanos** and E.A. Theodorou,
International Conference on Advanced Robotics (ICAR), 2023.
- [C3] [Decentralized Safe Multi-Agent Stochastic Optimal Control using Deep FBSDEs and ADMM](#),
M.A. Pereira*, **A.D. Saravanos***, O. So and E.A. Theodorou,
Robotics: Science and Systems (RSS), 2022.
- [C2] [Receding Horizon Differential Dynamic Programming Under Parametric Uncertainty](#),
Y. Aoyama, **A.D. Saravanos** and E.A. Theodorou,
IEEE Conference on Decision and Control (CDC), 2021.
- [C1] [Distributed Covariance Steering with Consensus ADMM for Stochastic Multi-Agent Systems](#),
A.D. Saravanos, A.G. Tsolovikos, E. Bakolas and E.A. Theodorou,
Robotics: Science and Systems (RSS), 2021.

Workshop Papers / Technical Reports

- [O3] [Bayesian Active Learning for Intent Disambiguation in Interactive Robot Planning](#),
H. Li, C. Sobolewski, **A.D. Saravanos**, W. Tan, J. Karigiannis and C. Fan
IROS Human-Robot Dialogue Workshop, 2026.
- [O2] [Sim2Real on the Robotarium Platform Using Decentralized Multi-Agent Safe Deep FBSDEs](#),
M.A. Pereira*, **A.D. Saravanos*** and E.A. Theodorou,
RSS Workshop on Scaling Robot Learning, 2022.
- [O1] [Sampling-Based Optimization for Multi-Agent Model Predictive Control](#),
Z. Wang, **A.D. Saravanos**, H. Almubarak, O. So and E.A. Theodorou,

Theses

- [T2] [Distributed Optimization Architectures for Large-Scale Decision-Making](#),
A.D. Saravanos
Ph.D. Thesis, Georgia Institute of Technology, 2025.
- [T1] [Nonlinear Model Predictive Control for Space Robotic Systems](#),
A.D. Saravanos
Diploma Thesis, University of Patras, 2019.

GRANTS & SPONSORED RESEARCH

- [G2] Multi-robot Systems with Human-Enabled Collaboration 2026–2027
GE Vernova
Award: \$300,000
Period: Oct 2026–Sep 2027
Role: Lead proposal author; PI: Chuchu Fan
- [G1] Safety Guarantees and Stability Verification for End-to-End Autonomous Driving 2026
Hyundai Motor Group (HMG)
Award: \$70,000
Period: May–Nov 2026
Role: Lead proposal author; PI: Chuchu Fan
- [S1] Autonomous Field Inspection with Physical AI 2025–2026
GE Vernova
Period: Oct 2025–Sep 2026
Role: Research lead; PI: Chuchu Fan
Led technical execution, sponsor presentations and successful on-site demonstration.

PATENTS

- [P3] [Collaborative learning with full model alignment](#) 2023
A.D. Saravanos, F.J. Cabrita Condessa, W.-Y. Lin, Z. Li and M.R. Ganesh
U.S. Patent Application No. 18/371,596.
- [P2] [Collaborative learning with full model alignment](#) 2023
A.D. Saravanos, F.J. Cabrita Condessa, W.-Y. Lin, Z. Li and M.R. Ganesh
U.S. Patent Application No. 18/371,594.
- [P1] [Collaborative learning with full model alignment](#) 2023
A.D. Saravanos, F.J. Cabrita Condessa, W.-Y. Lin, Z. Li and M.R. Ganesh
U.S. Patent Application No. 18/371,587.

AWARDS/FELLOWSHIPS

- Onassis Foundation Scholar - Doctoral Fellowship 2021-2025
Four-year award for PhD studies financial support
- Gerondelis Foundation - Doctoral Fellowship 2022
One-time award for PhD studies financial support
- Skouras Foundation Scholarship 2019
Top ECE student and in top 10 students of University of Patras for academic year 2018-19
- Valedictorian, Class of 2019, December Graduation Ceremony, ECE, University of Patras 2019
- 1st & 2nd places, Line Following Robots (Enhanced), Robotex International 2018, Estonia 2018
- 3rd place, Line Following Robots (Enhanced), Robotex International 2017, Estonia 2017
- Bronze medal, European Science Olympiad (EUSO) 2013, Luxembourg 2013
- Gold medal, National Science Olympiad 2013, Greece 2013

MENTORSHIP/SUPERVISION

Graduate Students

Carson Sobolewski, PhD Student at MIT	2025-present
<i>Research topic: Safe multi-agent control for autonomous field inspection robots</i>	
Arshiya Taj Abdul, MS/Junior PhD student at Georgia Tech	2023-2025
<i>Research topic: Distributed robust optimization for multi-agent control under uncertainty</i>	
<i>Resulted into 1 ACC '25, 1 CDC '25 and 1 TAC (under review) publications.</i>	
Panagiotis Theodoropoulos, Junior PhD student at Georgia Tech	2023-2025
<i>Research topic: Stochastic optimal control and generative AI</i>	
<i>Resulted into 1 ICLR '24 and 1 NeurIPS '25 publications.</i>	
Akash Ratheesh, Junior PhD student at Georgia Tech	2024-2025
<i>Research topic: Operator splitting covariance steering for safe stochastic nonlinear control</i>	
<i>Resulted into 1 CDC '25 publication.</i>	
Taehyun Yoon, Junior PhD student at Georgia Tech	2024-2025
<i>Research topic: Distributed trajectory optimization for multi-agent systems</i>	
<i>Resulted into an upcoming CDC '26 submission.</i>	
Joshua Kuperman, MS student at Georgia Tech	2022-2023
<i>Research topic: Tolerant barrier states for safe trajectory optimization</i>	
<i>Resulted into 1 ICAR '23 publication.</i>	

Undergraduate Students

Dami Thomas, UROP (Undergraduate Research Opportunities Program) at MIT	Fall 2025
<i>Research topic: Safe multi-agent control for autonomous field inspection robots</i>	
Hunter Kuperman, BS Student at Georgia Tech	Spring-Fall 2024
<i>Research topic: Learning-to-optimize for quadratic programming</i>	

TEACHING

AI in Robotics: Learning Algorithms, Design and Safety - MIT Professional Education	July 2026
<i>Co-instructor (with Prof. Chuchu Fan). Co-designed curriculum and delivered a 3-day intensive course for industry professionals on learning-based robotics, control, and safety.</i>	
Guest Lecturer at Feedback Control Systems 16.30/31, MIT	Dec 2025
<i>Lecture title: "Advanced Topics: Optimal Control and Lyapunov Analysis"</i>	
Guest Lecturer at Motion Planning RBE 550, Worcester Polytechnic Institute (WPI)	Nov 2025
<i>Lecture title: "Towards Large-Scale Autonomy: Multi-Agent Planning and Control"</i>	
Volunteer teacher, Robots at MET Program, University of Patras, Greece	2018-2020
<i>Taught teams of elementary school students on LEGO robotics</i>	
TISP (Teacher In-Service Program) Volunteer, IEEE Student Branch, University of Patras	2016-2020
<i>Taught elementary school students basic concepts in engineering</i>	
Volunteering course coordinator at "Skagiopouleio" Orphanage, Patras, Greece	May 2017
<i>Organized "Introduction to Coding" course for elementary school students</i>	

INVITED TALKS

2026 INFORMS Optimization Society Conference, Computational Optimization Track <i>Topic: Deep Learning-Aided Large-Scale Distributed Optimization</i>	Mar 2026
JST ASPIRE Meeting <i>Topic: Learning-to-Optimize via Generative Modeling</i>	Mar 2026
Worcester Polytechnic Institute - Robotics Engineering, Guest Lecture <i>Topic: Towards Large-Scale Autonomy: Multi-Agent Planning and Control</i>	Nov 2025

PEER REVIEW SERVICE

Journals: T-RO, TAC, TCNS, RA-L, L-CSS, IJRNC, TSMC.

Conferences: RSS, ICRA, IROS, ICLR, ICML, NeurIPS, L4DC, CDC, ACC, ECC, ICAR, IASEAI.

TECHNICAL SKILLS

Coding Languages:	Python, C/C++, MATLAB, Julia
Packages:	PyTorch, Scikit-learn, NumPy, CVX/CVXPY, MOSEK, Gurobi
IDEs:	Visual Studio, Jupyter Notebook, PyCharm, Spyder

LANGUAGES

English (Fluent), Greek (Native), French (Advanced)

REFERENCES

Prof. Evangelos A. Theodorou

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Dr. Wan-Yi Lin

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Pittsburgh, PA

 Google Scholar

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